

RESEARCH STATEMENT

My research interests are in program synthesis, formal methods, and symbolic logic. I am particularly interested in proofs as programs, the computational content of classical principles, and the relationship between syntax, semantics, proof search, and software construction. Before moving into research, I worked on reliable, air-gapped software systems at Google Cloud.

EDUCATION

Ashoka University

Bachelor of Science (Honours) in Computer Science

Major CGPA: 3.74/4

Sonipat, Haryana

Aug 2018 - Apr 2022

RELEVANT COURSEWORK

Discrete Mathematics; Algorithm Design and Analysis; Linear Algebra; Calculus; Abstract Algebra; Probability Theory; Programming Language Design and Implementation; Operating Systems; Computer Networks; Computer Security and Privacy.

EMPLOYMENT

Ashoka University

Research Associate · Programming Languages

· Studying semantics of programming languages, computational effects, and categorical approaches to programming-language foundations.

Sonipat, Haryana

Jun 2026 - Present

Firebolt

Software Engineer · Self-hosted Platform

· Joined the founding engineering team building Firebolt's first self-hosted platform for isolated, air-gapped enterprise deployments, including feature flags and Temporal workflow orchestration.

Bangalore, India

Nov 2025 - Mar 2026

Google

Software Engineer · Google Cloud, Distributed Cloud Team

· Built software for Google Distributed Cloud, bringing Google Cloud services into air-gapped, sovereign, and regulated environments through distributed systems that operated without public-network dependencies.

· Designed a Harbor-based artifact registry for a 40kg portable data center, adding Kubernetes self-healing, node-local fallback mirrors, and image signature verification to support offline bootstrap and operation.

· Built APIs and observability tooling for Prometheus/Grafana monitoring, covering system health, storage, image-pull latency, error rates, alerting, and dashboards for internal teams and deployment operators.

· Mentored junior engineers and hosted a summer intern who later joined full time.

Bangalore, India

Mar 2024 - Sept 2025

Google

Strategic Cloud Engineer · Google Cloud

· Helped enterprises become cloud-native on Google Cloud, combining technical consulting, hands-on implementation, and architecture guidance across cloud infrastructure projects.

· Researched and prototyped Data Insights, an LLM-powered natural-language layer for tabular data discovery, testing generated questions, SQL guardrails, accuracy, and latency for cold-start exploration of unfamiliar datasets; the capability later became part of a Google Cloud data management product.

· Built reusable cloud and data tooling, including Terraform modules for Kubernetes clusters and VPCs alongside open-source Dataproc templates for Spark/Hadoop ETL.

Bangalore, India

Jul 2022 - Mar 2024

TEACHING

AshokaX

Teaching Assistant · Machine Learning

· Assisted [Prof. Debayan Gupta](#) in teaching machine learning to high school students.

Online

Apr 2022 - Jun 2022

Ashoka University

Teaching Assistant · Computer Organization and Systems

· Assisted [Prof. Manu Awasthi](#) with Computer Organization and Systems in Ashoka's Computer Science department.

Sonipat, Haryana

Aug 2021 - Dec 2021

ACTIVITIES

· Participant, [Indian School on Logic and its Applications \(ISLA 2026\)](#), Krea University · Jun 25–30, 2026. Coursework: Programs and Proofs; Computability Theory.

· Participant, [Formal Methods Update Meeting 2026](#), Krea University · Jul 2–3, 2026.